You are the circuit fault team

Learning intention

Use a checked low-voltage circuit to compare electrical conductors and insulators.

A lamp is dark. Is the sample the problem?

Your job: check the evidence before classifying it.

Use a paper investigation or supervised low-voltage kit.

Find the missing idea

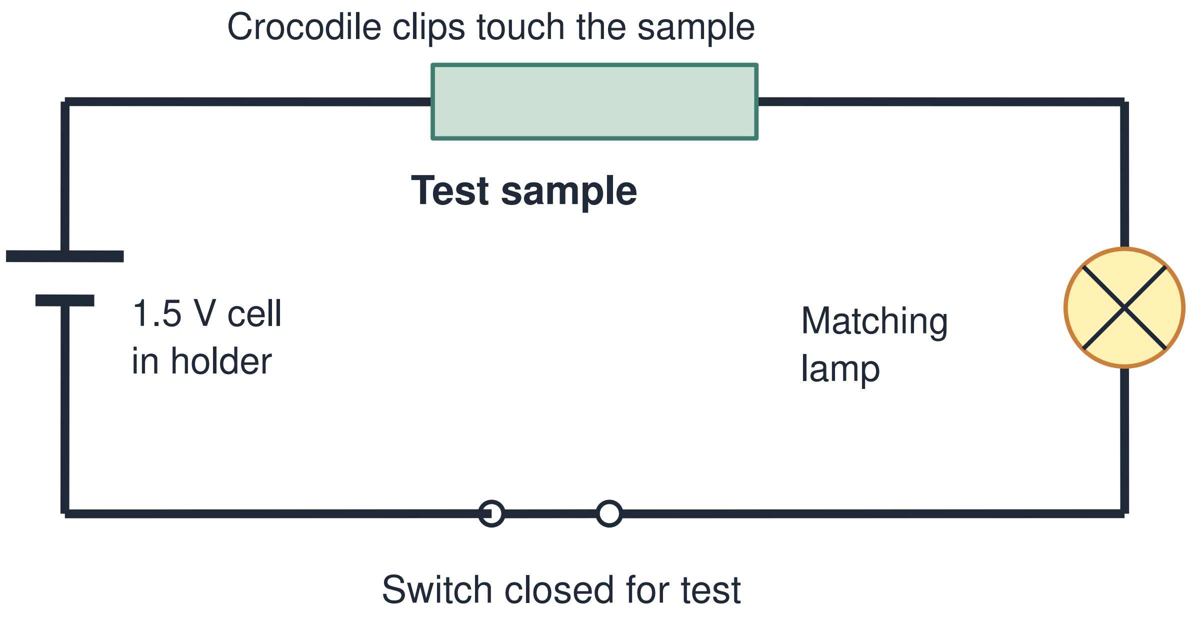
R1. Name a cell, lamp and wire.

R2. Does one wire from a cell to a lamp make a complete loop?

R3. Are thermal and electrical conductivity the same test?

Trace the whole loop

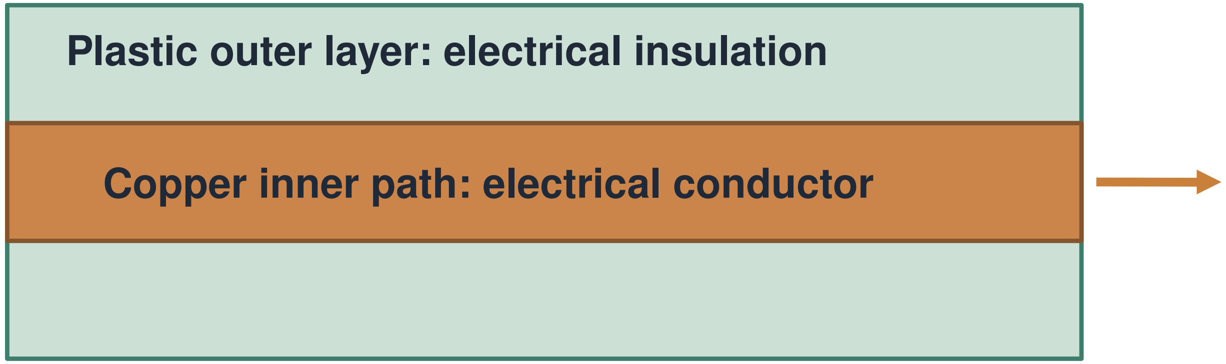
Every part must make a conducting connection



The sample bridges a gap in the circuit. The lamp stays in the loop. A good connection is needed at every join.

One lead needs two properties

Choose a material for its job

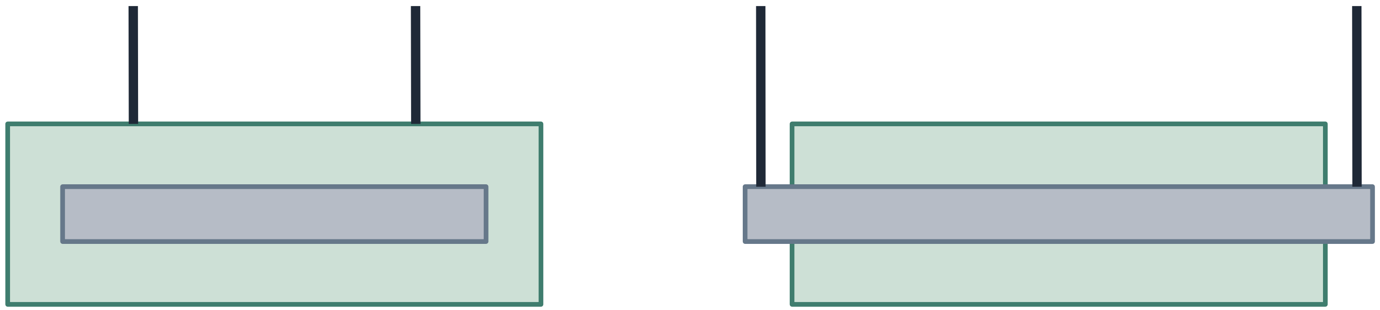


Cutaway model: sizes are enlarged.
Do not cut or open real electrical cables.

Copper provides a conducting path. Plastic insulates the outside. Both material choices serve a different purpose.

Do not trust a single dark lamp

Are the clips touching the material we mean?



Clips touch coating.
Metal core is not connected.
Result: uncertain for the metal.

Clips touch exposed metal.
Check with a known conductor.
Then interpret the lamp.

Use prepared samples only. No scraping or cutting.

Check copper before the sample. Check the sample contact, then observe. Check copper again afterwards.

Fault clinic: sort the case cards

Use the five cases on page 1.

Sort: conductor evidence / insulator evidence / uncertain.

Give the exact clue that supports your decision.

Repair the reasoning, not the sample

C: foil dark; copper control also dark.

D: coated metal dark; clips touch the coating.

What should we do before making a claim?

Build the class evidence table

Constructed observations from a checked 1.5 V circuit. Known copper lights before and after this sequence. Use “in this test” when classifying dark results.

Sample	Lamp observation	Contact check
Bare copper	Lights	Valid
Bare aluminium	Lights	Valid
Dry card	No visible glow	Valid
Rigid plastic	No visible glow	Valid

Constructed sample observations; copper controls passed before and after.

The control fails. What now?

A plastic sample gives no glow. The copper control also gives no glow.

A Plastic is definitely the cause.

B Check the circuit, then repeat the tests.

C Copper has become an insulator.

D Add many more cells to force a result.

Make your diagnostic report

Choose one response route with page 1 beside it.

Interpret the evidence, then design a two-layer lead.

State one result that must remain uncertain.

If using the optional kit

Copper check → sample → record → copper check.

Switch off for changes; keep the lamp in series.

Unexpected result? Mark uncertain and ask the teacher.

Use a scientific conclusion

“The lamp lit, which supports …”

“No glow with valid controls supports … in this test.”

“The control failed, so I cannot yet …”

Compare diagnoses

Which uncertain case changed your mind?

Explain why a good scientist may pause a conclusion.

Match conductor and insulation to their jobs.

Use our voice to improve the test

Choose the class's most useful next improvement.

Options: clearer contacts, repeated checks, clearer recording.

Explain how your choice improves the evidence.

Exit diagnosis

E1. Why must the circuit be a complete loop?

E2. Why check copper before and after a sample?

E3. Choose inside and outside materials for a lead.